# Forecasting Fantasy Premier League Points: A Walk-Forward Evaluation of Captaincy and Lineup Decisions

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**Abstract**

Fantasy Premier League (FPL) managers choose, every week, eleven starters from a fifteen-player squad and a captain whose points are doubled. We forecast each player's points for the coming gameweek with gradient-boosted regression trees and evaluate them on four seasons of data (2022/23–2025/26), with expanding training windows that contain only earlier weeks. Over 114 test gameweeks (2023/24–2025/26) and every registered player, the forecast is more accurate than recent form, a fixture-adjusted form rule and a ridge regression (forecasts exist for every player with at least one earlier club-fixture week). It also ranks realistic selection candidates better within each week. Its decisions are less clearly better. As a captain rule it outscores the most expensive attacker who played at least 60 minutes per fixture in his previous club-fixture week by +1.46 points per week (95% bootstrap interval +0.42 to +2.49), almost entirely because of one season, and unadjusted recent form by +1.37. Its advantages over a form rule scaled by the coming week's fixture count and over a ridge regression are not distinguishable from zero. It trails the captain most often chosen by the managers who finished each season in the overall top 100 by −1.59 (−2.50 to −0.73); that cohort is defined after the season, so this is a retrospective reference. The model does not meet a decision criterion recorded before the first evaluation. A football adaptation of basketball's usage rate reduced forecast error by about 0.3% and had no detectable effect on decisions. Performance varies markedly between seasons, and resolving captaincy differences of a point per week or less needs several seasons of prospective evaluation.

## 1 Introduction

Fantasy Premier League is the official fantasy game of the English Premier League. Each manager holds fifteen players within a fixed budget and, before each of the season's 38 gameweek deadlines, names eleven starters and a captain whose points are doubled (Fantasy Premier League, 2026b). Players score for playing time, goals, assists, clean sheets, saves and bonus points and, from 2025/26, for defensive actions. Two decisions recur every week regardless of transfers: who starts, and who captains. Both reduce to ranking players by expected points in the coming week.

A small literature forecasts FPL points. Early work treated team selection as sequential decision-making under uncertainty (Matthews et al., 2012). Later studies forecast points from recent match histories using time-series and recurrent models (Gupta, 2019), convolutional and recurrent networks over a player's last five matches (Lombu et al., 2024), and ridge regression, gradient boosting and convolutional networks (Frees et al., 2024). Others add news and betting-market information (Bonello et al., 2019) or couple forecasts with mixed-integer squad selection (Ramezani and Dinh, 2025). These studies mostly report forecast error or season-long team totals. We ask whether a forecast makes better *weekly decisions* than simple rules and expert consensus, judged against a criterion fixed before the results were seen.

Two issues receive particular attention: information leakage, meaning the use of anything not known at decision time (Tashman, 2000), and the noise of weekly outcomes, which limits what any evaluation can show. Our contributions are:

1. a deadline-ordered feature set that carries players' form across seasons;
2. a walk-forward evaluation that retrains every week on every registered player, and compares decisions with mechanical, statistical and expert comparators;
3. lineup evaluation on 11,400 real squads, split into the eleven and the armband;
4. ablation, sensitivity and dependence analyses; and
5. a test of whether ball usage, adapted from basketball (Basketball-Reference, 2026; Oliver, 2004), adds information.

# 2 Data

## 2.1 Player-gameweek records

We use official FPL data for four complete seasons, 2022/23 to 2025/26 (Fantasy Premier League, 2026a). These weekly records were collected by the author during each season. Fixture lists and club identifiers for past seasons come from a public archive of the same source (Anand, 2026). For every registered player and gameweek we record:

- minutes, starts, goals, assists, clean sheets, goals conceded and saves;
- bonus points and the bonus points system score;
- expected goals (xG), expected assists (xA) and expected goals conceded;
- total FPL points.

Expected goals estimate the probability that a shot is scored given its characteristics (Rathke, 2017); FPL publishes Opta's values. Double gameweeks, in which a club plays twice, are summed. Players and clubs keep the same identifier across seasons.

Every registered player is kept, including those who never played that season (Table 1). Whether a player will play is not known at earlier deadlines, so removing those who end the season without minutes would select the evaluated population with hindsight.

**Table 1:** Player-gameweek records by season (every registered player).

| Season | Player-gameweeks | Players | Total FPL points |
|---|---|---|---|
| 2022/23 | 26,337 | 778 | 31,724 |
| 2023/24 | 29,679 | 865 | 31,271 |
| 2024/25 | 27,279 | 784 | 31,291 |
| 2025/26 | 29,747 | 841 | 34,382 |
| Total | 113,042 | | |

Weekly prices were not archived, so each player carries his start-of-season price. The 2025/26 scoring rules added points for defensive actions (Fantasy Premier League, 2026b), which partly explains that season's higher total; a season indicator lets the model treat it separately.

**Fixtures.** Fixture assignments are the final ones. When a match is postponed, its original gameweek becomes a blank for both clubs and its new gameweek may become a double. At the original deadline the match was still expected, so final assignments contain some hindsight about which clubs play. We cannot reconstruct the schedule as it stood at each deadline. Instead, we report results separately for the 91 test weeks in which every club played exactly once and the 23 weeks with a blank or double gameweek (Table 5). We also assume that a gameweek's outcomes were known by the next deadline, which holds for completed gameweeks in FPL's schedule.

### 2.2 Squads of top-ranked managers

For 2023/24 to 2025/26 we hold the weekly fifteen-player squads, starting elevens, bench order, captain and vice-captain of the 100 managers who finished each season in the overall top 100: $3 \times 38 \times 100 = 11{,}400$ squad-weeks. Membership of this cohort is determined by the end of the season, so the cohort is selected on the outcome (Brown et al., 1992). We use its most-captained player as a retrospective reference and its squads as a realistic test bed. Lineup results are therefore conditional on these managers' squads.

### 2.3 Season event data

For the usage analysis (Section 5) we use Opta season totals for every Premier League player from 2021/22 to 2025/26, published through the Premier League's statistics services (Premier League, 2026). Each player is assigned to the club whose season squad lists him or, for players who left mid-season, to the club where he played most of his minutes.

## 3 Method

### 3.1 Notation and information ordering

Index players by $i$, seasons by $s$ and gameweeks by $g$, and write $t = (s, g)$ for a point in time ordered lexicographically. Let $y_{it}$ be player $i$'s points in week $t$, $m_{it}$ his minutes and $n_{it} \in \{0, 1, 2\}$ the number of fixtures his club plays. Features for week $t$ are computed from weeks $t' < t$ and from quantities fixed before the deadline: position, start-of-season price, previous-season totals and the fixture list (subject to the caveat above). We check the ordering by perturbing the outcomes of one week and confirming that features for that week and all earlier weeks are unchanged.

### 3.2 Features

A player's state is updated after each gameweek in which his club had a fixture (a *club-fixture week*); the features for week $t$ take the latest state from a strictly earlier week. Table 2 lists the design choices with their basis.

**Form and role.** Minutes per fixture in the last club-fixture week and averaged over the last three and six; the average, over the last six, of starts per fixture; and points per fixture averaged over the last three and six. Rolling windows of recent matches are the common input in FPL forecasting (Frees et al., 2024; Lombu et al., 2024; Ramezani and Dinh, 2025). Windows run across season boundaries.

**Per-90 rates.** Over the last six club-fixture weeks $\mathcal{W}_6$, for a statistic $z$ (xG, xA, bonus, bonus points system score, saves),

$$r_{it}^{(6)}(z) = 90 \cdot \frac{\sum_{t' \in \mathcal{W}_6} z_{it'}}{\max\left(\sum_{t' \in \mathcal{W}_6} m_{it'},\, 90\right)},$$

where the floor of 90 minutes stops a short cameo from producing an extreme rate.

**Season to date.** The same rates for xG, xA and points accumulated over the current season, the share of available minutes played, $\sum m / (90 \sum n)$, and the number of fixtures so far. In a season's first week the rates are missing and the fixture count is zero.

**Prior season.** The player's xG, xA and points per 90, and his share of the season's minutes, in the previous season. The motivation is by analogy with empirical Bayes estimation, which improves noisy early-season rates by combining them with a longer-run prior (Brown, 2008; Efron and Morris, 1975). The model is not an empirical Bayes estimator; the trees decide how to use these inputs, and Section 4.4 tests whether they help.

**Club and opponent strength.** For club $c$ in week $t$, let $G^+_{ct} = \sum_{i \in c} \mathrm{xG}_{it}$. As a proxy for xG conceded we take $G^-_{ct} = \sum_o G^+_{ot}/n_{ot}$ over its opponents $o$ that week. In a double gameweek this splits an opponent's weekly xG equally across its fixtures, an approximation forced by weekly rather than match-level data. Over the club's previous eight club-fixture weeks, which can span two seasons,

$$A_{ct} = \frac{\sum_{t'} G^+_{ct'}}{\sum_{t'} n_{ct'}}, \qquad D_{ct} = \frac{\sum_{t'} G^-_{ct'}}{\sum_{t'} n_{ct'}}.$$

Separate attacking and defensive strengths per club are the basis of the standard goal models for football (Dixon and Coles, 1997; Maher, 1982). We build them from xG because goals are rare and xG summarises the quality of the chances a team creates and concedes (Rathke, 2017). The player receives his club's $A$ and $D$, his opponents' average $A$ and $D$ in week $t$, his number of fixtures and the share played at home.

**Static and exposure.** Position, start-of-season price, gameweek number, an indicator for the defensive-action scoring rules, the number of earlier weeks in which he played, and the number of gameweeks since his last club-fixture week.

**Table 2:** Design choices and their basis.

| Choice | Basis |
|---|---|
| Previous-season rates | By analogy with empirical Bayes shrinkage (Brown, 2008; Efron and Morris, 1975); tested by ablation (Table 7) |
| Rolling windows of recent matches | Common in FPL forecasting (Frees et al., 2024; Lombu et al., 2024; Ramezani and Dinh, 2025) |
| Club attack and defence strength | Team attack and defence parameters in goal models (Dixon and Coles, 1997; Maher, 1982); xG as a summary of chance quality (Rathke, 2017) |
| Gradient-boosted trees | Friedman (2001); histogram-based variant as popularised by Ke et al. (2017); boosting used for FPL by Frees et al. (2024) |
| Model settings | Fixed before evaluation; sensitivity in Table 7 |
| Rolling-origin evaluation | Tashman (2000); on cross-validation with dependent data, Bergmeir and Benítez (2012) |
| Paired comparison on common outcomes | Diebold and Mariano (1995); bootstrap (Efron and Tibshirani, 1993; Künsch, 1989; Politis and Romano, 1994) |
| Decision criterion recorded in advance | In the spirit of pre-registration (Nosek et al., 2018); not formally registered |
| Usage rate | Basketball usage percentage (Basketball-Reference, 2026; Oliver, 2004); football adaptation (Sincere FC, 2025a,b) |

## 3.3 Model

The target is $y_{it}$. We fit histogram-based gradient-boosted regression trees (Friedman, 2001), in scikit-learn's implementation (Pedregosa et al., 2011), which follows the histogram approach of Ke et al. (2017); no LightGBM model was fitted. The loss is squared error. The settings are:

- learning rate 0.05 and at most 250 boosting iterations, with the library's default early stopping above 10,000 training rows (training stops when the loss on a random 10% of the training window fails to improve for ten iterations);

- at most 31 leaves per tree and at least 80 observations per leaf;
- an $L_2$ penalty of 1.0.

They start from the library defaults (learning rate 0.1, 100 iterations, 20 observations per leaf, no penalty) and move each setting in the conservative direction suited to a noisy target. In the 114 weekly fits, early stopping ended training after 70 to 175 iterations (median 114.5). The internal validation split uses only the training window, so it does not leak information from the test week. Missing inputs are handled natively by the trees. Features that are constant or entirely missing within a training window are dropped for that fit. A player with no earlier club-fixture week receives no forecast. Training uses every player-week with a fixture and at least one earlier club-fixture week.

### 3.4 Decision rules

**Captain.** The model captains the player with the highest forecast among those whose club has a fixture. The captain endpoint is the nominated player's own points that week, single and without vice-captain fallback.

**Lineup.** Given a fifteen-player squad, the model chooses the legal formation that maximises the sum of forecasts. A formation has one goalkeeper, $d$ defenders, $m$ midfielders and $f$ forwards, with $d \in \{3, 4, 5\}$, $m \in \{2, \ldots, 5\}$, $f \in \{1, 2, 3\}$ and $d + m + f = 10$. The two highest forecasts in the eleven are captain and vice-captain. The bench lists the reserve goalkeeper first and the outfield players by forecast. Lineups are scored by the game's ordinary rules (Fantasy Premier League, 2026b):

- a starter with no minutes is replaced by the first bench player, in bench order, who played and whose entry keeps the formation legal;
- the captain's points are doubled, or the vice-captain's if the captain did not play;
- chips are ignored (a triple captain counts as an ordinary captain, a bench boost as an ordinary week).

### 3.5 Comparators

**Premium attacker.** The midfielder or forward with the highest start-of-season price among those who played at least 60 minutes per fixture in their previous club-fixture week. Exact price ties are broken by points per fixture over the last six, then by player identifier.

**Recent form.** The highest points per fixture over the last six, among players averaging at least 60 minutes per fixture over the last three.

**Top-100 consensus.** The player most often captained that week by the managers who finished the season in the overall top 100 (ties to the lower identifier). Not available at the deadline; a retrospective reference.

Two further comparators were added after an independent review, so they are not part of the decision criterion. *Recent form* × *fixtures* multiplies recent form by the number of fixtures in the coming week. *Ridge regression* is a walk-forward linear model on start-of-season price, recent minutes, position and the number of fixtures. For lineups, the comparators are the managers' own eleven, bench order and captain, and the model's selection procedure applied to each comparator's ranking.

### 3.6 Evaluation

For each test week $w$ the model is retrained on all weeks before $w$, including earlier weeks of the same season, and used to forecast week $w$. This rolling-origin design mirrors live use (Tashman, 2000). Random cross-validation can misstate accuracy when observations are dependent over time (Bergmeir and Benítez, 2012); here, rolling form and the season structure create exactly that dependence. The test period is the 114 gameweeks of 2023/24, 2024/25 and 2025/26.

Comparisons are paired by week, so rules are judged on the same outcomes (Diebold and Mariano, 1995). For rule $A$ and comparator $B$, let $d_w = \text{pts}_w(A) - \text{pts}_w(B)$; for lineups, $\text{pts}_w$ is first averaged over the 100 squads in week $w$. The primary interval resamples the 114 weekly differences with replacement (Efron and Tibshirani, 1993), 10,000 times. Because consecutive weeks may be dependent, we also report a moving-block bootstrap with blocks of six weeks that never cross a season (Künsch, 1989; Politis and Romano, 1994), the lag-one autocorrelation of $d_w$, and season-by-season estimates. Inference refers to these three seasons; extrapolating to future seasons assumes the game and the data stay comparable.

**Decision criterion.** Written into the evaluation code before the first run, and recorded in that code's first version-controlled commit: the model's captain choice meets the criterion only if the lower end of the interval is above zero against *every* comparator in the original set (premium attacker, recent form, top-100 consensus). This is an intersection–union rule, so no multiplicity adjustment is needed. The same rule applies to lineups. The study was not formally registered, so this pre-specification is reported by the author.

**Forecast accuracy.** We report the mean absolute error (MAE), root mean squared error and Pearson correlation over all test player-weeks with a fixture. We also report the mean, over weeks, of the within-week Spearman rank correlation, and the same measures among *regulars* (players averaging at least 60 minutes per fixture over the last three), who are the realistic selection candidates.

## 4 Results

### 4.1 Forecast accuracy

Table 3 compares the model with the three forecasting comparators over 84,200 test player-weeks. The model has the lowest error and the highest correlation on every measure. Its within-week rank correlation is 0.71, against 0.69 for form × fixtures and 0.68 for the ridge regression. Among regulars, the players a manager actually chooses between, the gap is wider: 0.31 against 0.21 and 0.28. Paired by week, the model's MAE is lower than form × fixtures by +0.057 (+0.048 to +0.066) and lower than the ridge regression by +0.048 (+0.041 to +0.056). A pooled linear calibration check, regressing outcomes on forecasts, gives a slope of 0.97 and an intercept of 0.01. The rank correlations are point estimates; we did not compute intervals for them.

**Table 3:** Forecast accuracy over all test player-weeks with a fixture, and among regulars (20,400 player-weeks). Weekly rank is the mean within-week Spearman correlation.

| | MAE | RMSE | Pearson | Weekly rank | MAE, regulars | Weekly rank, regulars |
|---|---|---|---|---|---|---|
| Model | 0.996 | 1.943 | 0.575 | 0.71 | 2.286 | 0.31 |
| Recent form | 1.052 | 2.109 | 0.493 | 0.69 | 2.480 | 0.20 |
| Recent form × fixtures | 1.053 | 2.102 | 0.504 | 0.69 | 2.477 | 0.21 |
| Ridge regression | 1.046 | 1.987 | 0.547 | 0.68 | 2.341 | 0.28 |

## 4.2 Captaincy

Table 4 and Figure 1 give the captain results. The model's captains average 6.61 points per week.

- It outscores the premium-attacker rule by +1.46 (+0.42 to +2.49) and unadjusted recent form by +1.37 (+0.22 to +2.48).
- Against the comparators added after review the differences are not distinguishable from zero: +0.43 (−0.81 to +1.62) against form × fixtures and +0.87 (−0.21 to +1.93) against the ridge regression. Much of the advantage over unadjusted form therefore comes from knowing which players have two fixtures.
- It trails the top-100 consensus by −1.59 (−2.50 to −0.73), so it does not meet the decision criterion.

The best possible captain each week averaged 17.79 points, a mean regret of 11.18.

**Table 4:** Captain points per gameweek (single) by season.

| Season | Model | Top-100 consensus | Premium attacker | Recent form |
|---|---|---|---|---|
| 2023/24 | 5.89 | 7.29 | 5.92 | 4.68 |
| 2024/25 | 9.11 | 10.13 | 5.26 | 6.71 |
| 2025/26 | 4.84 | 7.18 | 4.29 | 4.34 |
| All (114 weeks) | 6.61 | 8.20 | 5.16 | 5.25 |

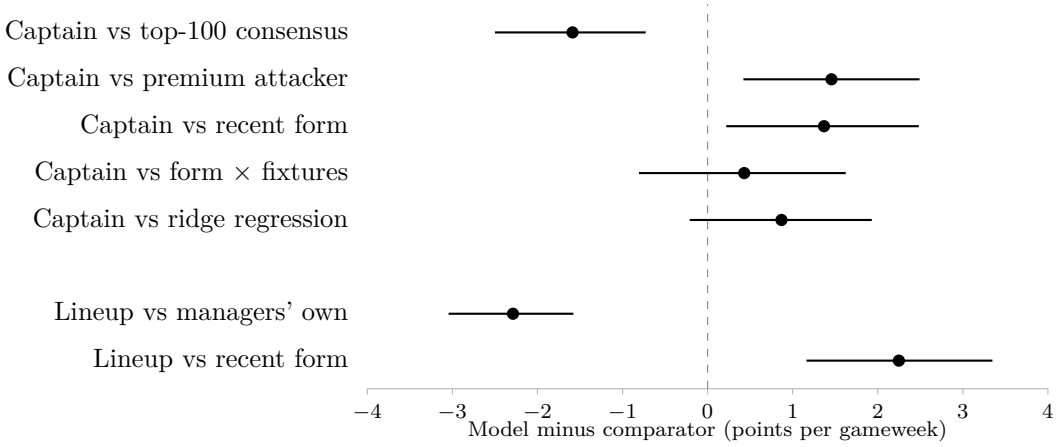



**Figure 1:** Paired weekly differences with 95% bootstrap intervals over 114 gameweeks. Right of zero favours the model. Form × fixtures and the ridge regression were added after review.

Table 5 examines the inference.

- *Seasons differ.* The advantage over the premium rule is −0.03, +3.84 and +0.55 points per week in 2023/24, 2024/25 and 2025/26, so it rests almost entirely on 2024/25. The deficit to the top-100 consensus appears in every season (−1.40, −1.03, −2.34).
- *Serial dependence.* Averaged over seasons, the lag-one autocorrelations of the weekly differences lie between −0.11 and +0.14; for individual comparator–season pairs they range from −0.23 to +0.28. Three- and six-week block bootstraps within seasons give intervals close to the independent resampling. Lag-one correlations alone do not rule out longer-range dependence, and the lineup, accuracy and usage intervals use independent weeks only.

- *Regular weeks.* Over the 91 weeks in which every club played once, the mean advantages are +1.41 over the premium rule, +0.80 over recent form, +0.80 over form × fixtures and +1.05 over the ridge regression, against −1.58 for the top-100 consensus. These point estimates resemble the full results, but regularity is itself judged from final fixture assignments, and training data still carry any schedule hindsight, so this does not remove the limitation.

**Table 5:** Captain comparisons under alternative inference. Columns: comparator points per week; the model's mean advantage with the independent-week interval; the within-season six-week block bootstrap interval; season means; and the mean over regular weeks (every club playing once). [a]Blocks never cross a season. [b]Cohort defined after the season; retrospective reference.

| Comparator | Pts/wk | Model minus comparator (95% CI) | Block CI[a] | 23/24 | 24/25 | 25/26 | Regular wks |
|---|---|---|---|---|---|---|---|
| Top-100 consensus[b] | 8.20 | −1.59 (−2.50 to −0.73) | −2.26 to −0.55 | −1.40 | −1.03 | −2.34 | −1.58 |
| Premium attacker | 5.16 | +1.46 (+0.42 to +2.49) | +0.73 to +2.51 | −0.03 | +3.84 | +0.55 | +1.41 |
| Recent form | 5.25 | +1.37 (+0.22 to +2.48) | +0.37 to +2.09 | +1.21 | +2.40 | +0.50 | +0.80 |
| Recent form × fixtures | 6.18 | +0.43 (−0.81 to +1.62) | −0.82 to +1.39 | +0.13 | +1.16 | 0.00 | +0.80 |
| Ridge regression | 5.75 | +0.87 (−0.21 to +1.93) | −0.15 to +2.06 | −0.16 | +2.84 | −0.08 | +1.05 |

## 4.3 Lineups

On the 11,400 squad-weeks, the model's choices score 66.00 points per squad-week, against 68.29 for the managers' own choices and 63.76 for the form rule. The model beats the form rule by +2.25 (+1.16 to +3.35) and trails the managers by −2.29 (−3.04 to −1.58). Table 6 splits these totals into the eleven (after substitutions) and the armband.

- The managers' advantage comes from both parts: −1.28 (−1.76 to −0.86) in the eleven and −1.01 (−1.67 to −0.42) from the armband.
- Against form × fixtures the model gains +1.61 (+0.44 to +2.75) in total.
- Against the ridge regression there is no detectable difference in total (−0.14 (−0.98 to +0.65)). The model picks the better eleven (+0.61 (+0.17 to +1.06)), and the regression the better captain (−0.75 (−1.49 to −0.08)).

**Table 6:** Lineup points per squad-week, split into the eleven and the armband, and the model's paired differences with 95% intervals.

| | Eleven | Armband | Model minus: eleven (95% CI) | armband (95% CI) |
|---|---|---|---|---|
| Model | 58.74 | 7.26 | | |
| Managers' own choices | 60.02 | 8.28 | −1.28 (−1.76 to −0.86) | −1.01 (−1.67 to −0.42) |
| Recent form | 57.19 | 6.57 | +1.55 (+0.95 to +2.16) | +0.70 (−0.23 to +1.65) |
| Recent form × fixtures | 57.42 | 6.97 | +1.31 (+0.72 to +1.92) | +0.29 (−0.72 to +1.28) |
| Ridge regression | 58.13 | 8.01 | +0.61 (+0.17 to +1.06) | −0.75 (−1.49 to −0.08) |

## 4.4 Ablations and settings

Table 7 repeats the evaluation with one change at a time: a block of features removed, or one model setting moved, giving 10 variants plus the reference. Ablation shows what each block contributes *to each endpoint* given the others. Correlated blocks can substitute for one another, so the changes are not intrinsic importances.

- *Accuracy.* Removing minutes and starts raises MAE from 0.996 to 1.014 and lowers correlation from 0.575 to 0.560. Removing club and opponent strength gives 1.005 and 0.571, and removing season-to-date rates 1.004 and 0.572. Removing last season's rates gives 0.996 and 0.576.

- *Captain and lineup points* change between variants, not always in the direction of accuracy. We did not estimate paired intervals for these changes, so they should not be read as differences between variants.
- *The verdict.* Across all variants MAE stays within 0.018 and correlation within 0.015 of the reference. Every variant beats the premium-attacker rule (lowest lower bound +0.12) and trails the top-100 consensus (highest upper bound −0.19). The advantage over recent form is less robust: in 1 of the 10 variants plus the reference its interval includes zero.

**Table 7:** Ablations and settings. MAE and correlation over test player-weeks; captain points per week; the captain's mean weekly difference from the top-100 and premium comparators; and the lower end of its interval against recent form.

| | MAE | Corr. | Captain | vs top-100 | vs premium | vs form, lower CI |
|---|---|---|---|---|---|---|
| Reference model | 0.996 | 0.575 | 6.61 | −1.59 | +1.46 | +0.22 |
| Without prior season | 0.996 | 0.576 | 6.89 | −1.32 | +1.73 | +0.44 |
| Without club and opponent strength | 1.005 | 0.571 | 6.91 | −1.29 | +1.75 | +0.52 |
| Without per-90 rates (last six) | 0.996 | 0.574 | 6.51 | −1.69 | +1.35 | +0.13 |
| Without season to date | 1.004 | 0.572 | 6.79 | −1.41 | +1.63 | +0.33 |
| Without minutes and starts | 1.014 | 0.560 | 6.33 | −1.88 | +1.17 | −0.10 |
| Learning rate 0.1, 125 iterations | 0.995 | 0.574 | 6.71 | −1.49 | +1.55 | +0.18 |
| 15 leaves per tree | 0.993 | 0.577 | 6.76 | −1.44 | +1.60 | +0.35 |
| 63 leaves per tree | 0.997 | 0.574 | 7.14 | −1.06 | +1.98 | +0.82 |
| 20 observations per leaf | 0.995 | 0.575 | 6.96 | −1.24 | +1.81 | +0.72 |
| Library defaults | 0.998 | 0.572 | 6.99 | −1.21 | +1.83 | +0.69 |

# 5 Extension: ball usage

## 5.1 Motivation and definition

In basketball, usage percentage estimates the share of a team's possessions a player ends with a shot, free throws or a turnover while on the floor (Basketball-Reference, 2026), building on the possession framework of Oliver (2004). A football adaptation (Sincere FC, 2025a,b) counts *possession-ending actions*: shots, and every way of losing the ball. It argues that the measure describes a player's role within a team, and that a player used to being the main option may struggle at a club that already has one. If so, last season's usage could inform this season's output, particularly after a transfer.

We count a player's actions as Opta's possessions lost plus his shots. A shot does not always end a possession, so this is an approximation. Possessions lost covers incomplete passes and crosses, failed take-ons, being tackled and miscontrols, and more: for Premier League players with at least 1,500 minutes in 2023/24–2025/26, it is a median 8–9% larger than the sum of those itemised components, depending on the season (the seasons' 5th and 95th percentiles span 1–31% larger). We then compute a minutes-adjusted share of the team's recorded actions, following the basketball formula with eleven players in place of five:

$$\mathrm{USG}_i = 100 \cdot \frac{\mathrm{PEA}_i \cdot (\text{team minutes}/11)}{\text{minutes}_i \cdot \mathrm{PEA}_{\text{team}}}.$$

Team totals sum every player of that club with recorded actions. Season totals cannot isolate the team's actions while the player was on the pitch. A player who changed clubs mid-season has all his season's actions assigned to one club. For creation we use Opta's chances created (passes leading directly to a shot); the adaptation's broader shot-creating actions are not available to us.

### 5.2 Features and test

Six features summarise the previous season:

- usage rate;
- possession-ending actions per 90;
- chances created per 90 and per action;
- touches in the opposition box per 90;
- club possession share.

Two describe the player's current context at the deadline: an indicator that his current club differs from the club he ended last season with, and the highest previous-season usage among his current teammates (excluding himself). We rerun the walk-forward evaluation with and without these eight features on identical weeks.

**Table 8:** Walk-forward evaluation with and without the usage features (2023/24–2025/26), paired by gameweek with 95% bootstrap intervals. [a]Mean over weeks of the reduction in weekly MAE (positive means the usage features help); the MAE columns are pooled over player-weeks.

| | Without usage | With usage | Difference (95% CI) |
|---|---|---|---|
| Mean absolute error[a] | 0.996 | 0.993 | +0.003 (+0.001 to +0.004) |
| Correlation | 0.575 | 0.576 | |
| Captain points per week | 6.61 | 6.92 | +0.31 (−0.24 to +0.85) |
| Lineup points per week | 66.00 | 66.05 | +0.04 (−0.35 to +0.43) |
| MAE, players who changed club[a] | | | +0.002 (−0.003 to +0.007) |

### 5.3 Result

Coverage is measured on the usage rate itself, over all 84,995 test player-weeks with a fixture (including the 795 without earlier history, which receive no forecast). The usage rate is present for 46% of these player-weeks and 67% of regulars' player-weeks. Of all test player-weeks, 45% belong to players with no Premier League minutes the season before. A further 9% belong to players who had minutes but no matched or complete usage record. Players who had changed club account for 9% of test player-weeks.

- *Accuracy.* Weekly MAE falls by +0.003 (+0.001 to +0.004): detectable, but about 0.3% of the error. For players who changed club the change is +0.002 (−0.003 to +0.007).
- *Decisions.* Captain points change by +0.31 (−0.24 to +0.85) and lineup points by +0.04 (−0.35 to +0.43), neither distinguishable from zero.

In this implementation and sample, last season's usage adds a small amount of information about points and no detectable value for decisions. The intervals do not exclude captain gains of up to +0.85 points per week. A match-level, in-season measure, or a club-level model of how a signing redistributes the ball, may behave differently.

## 6 Discussion

**What the evidence supports.** The forecast is more accurate than every forecasting comparator, ranks regulars better within each week, and passes a pooled linear calibration check. Better forecasts do not yet translate into clearly better decisions. As a captain rule the model beats

a pure price rule and unadjusted form, but not by a detectable margin once the comparator accounts for double gameweeks or is itself a simple regression. In lineups it beats both form rules, including the fixture-scaled one, and shows no detectable difference from the regression. The undetected captain differences are under a point per week, which three seasons resolve poorly (see the precision and power analysis below).

**The top-100 comparison.** The model trails the top-100 consensus. The data cannot say how much of that gap is information the model lacks, how much is skill, and how much is selection. The cohort is chosen after the season, so its choices are, on average, ones that worked out (Brown et al., 1992). At the deadline managers also knew team news, late injuries and penalty duty; earlier work found that adding news and betting-market information improves FPL forecasts (Bonello et al., 2019). A cohort fixed before the season, such as the previous season's top managers, would be the fair deployable benchmark; we did not have its weekly squads.

**Precision and power.** The standard deviation of the weekly captain difference ranges from 4.9 to 6.6 points across comparators. Taking $\sigma = 5$ for illustration and assuming independent weeks:

- A 95% interval of half-width $h$ needs about $(1.96\,\sigma/h)^2$ weeks: 96 weeks for $h = 1$, 384 for $h = 0.5$. One season gives $h \approx 1.6$.
- Detecting a true difference $\delta$ with 80% power in a two-sided 5% test needs about $\big((1.96 + 0.84)\,\sigma/\delta\big)^2$ weeks: 196 weeks (about five seasons) for $\delta = 1$ and 784 (about twenty) for $\delta = 0.5$.

A season of results therefore says little about captaincy rules that differ by a point a week or less, and larger $\sigma$ makes this worse. Differences between variants of the same model need their own paired estimates, which Table 7 does not provide.

**Limitations.**

- *Information.* No team news, injury history, set-piece duties or bookmaker prices.
- *Schedules.* Final fixture assignments contain some hindsight about postponements; regular-week results are reported for comparison.
- *Prices.* Only start-of-season prices are available historically.
- *Outcome-selected cohort.* The top-100 comparisons are retrospective.
- *Proxies.* Weekly rather than match-level data force the xG-conceded allocation and season-total usage measures described above.
- *Reuse.* The usage extension and the added comparators were evaluated on the same seasons as the main analysis; confirmation should be prospective.
- *Usage measurement.* We did not check the completeness of team denominators or the sensitivity of usage to minimum-minute thresholds.

**Prospective protocol.** From gameweek 7 of 2026/27, three captains are logged before each deadline and frozen at it:

- the model's;
- the one with the highest official expected-points figure;

- a live premium rule.

The live premium rule differs from the historical one. It uses the current price, a start in the previous calendar gameweek and current form, because that information is available live. The most-captained player across all managers is recorded once the gameweek begins. Official expected points, ownership and availability flags are archived at each deadline.

# 7 Conclusion

Evaluated week by week over three seasons on every registered player, a gradient-boosted forecast predicts and ranks players better than form-based and regression comparators. Its captain choices beat naive rules but are not distinguishable from those of a fixture-aware form rule or a simple regression. Its lineups beat both form rules and are not distinguishable from the regression's. Both fall short of the choices of each season's eventual top 100 managers. Distinguishing rules that differ by a point a week or less needs several seasons of evidence, so further evaluation should be prospective and specified in advance.

**Reproducibility.** All results were produced in one software environment (Python 3.11, scikit-learn 1.9.1, pandas 3.0.6, NumPy 2.4.6) from the analysis scripts accompanying this paper. The numbers in the results sections, tables and figures are generated from the result files by script; Table 1 and the illustrative precision and power calculations are written by hand. Weekly captain selections and outcomes are archived with the scripts; player-level forecasts and weekly lineup differences are not.

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# A Features

**Table 9:** Model inputs. A club-fixture week is a gameweek in which the player's club played; all windows end strictly before the forecast week. The usage block is used only in Section 5.

| Group | Features |
|---|---|
| Static | position; start-of-season price; gameweek number; defensive-action scoring era (0/1) |
| Fixture | number of fixtures in the week; share at home; own club's and opponents' rolling xG for and xG-conceded proxy (previous eight club-fixture weeks) |
| Form | minutes per fixture (last 1, mean of last 3, mean of last 6); mean starts per fixture (last 6); points per fixture (last 3, last 6) |
| Per-90 rates (last 6) | xG, xA, bonus points, bonus points system score, saves |
| Season to date | xG, xA and points per 90; share of available minutes played; fixtures so far |
| Prior season | xG, xA and points per 90; share of the season's minutes |
| Exposure | number of earlier weeks with minutes; gameweeks since the last club-fixture week |
| Usage (extension) | previous season's usage rate, possession-ending actions per 90, chances created per 90 and per action, box touches per 90, club possession; changed club (0/1); highest previous-season usage among current teammates |